



Lect. 13 Super-Resolution for Remote Sensing Image (RSI)

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Lect.13 Super-Resolution for Remote Sensing Image (RSI)

- Review of RSI Super-Resolution (RSISR)**
- Traditional Methods for RSISR**
- Deep Learning Methods for RSISR**
- Performance Evaluation Methods**
- Future Prospects**
- Cases**



Review of RSI Super-Resolution

- **Definition and Types of RSI Super-Resolution (RSISR)**
- **Why is RSISR important?**
- **Application Scenarios of RSISR**
- **Open Source Datasets for RSISR**



Definition and Types of RSISR

➤ Definition of Image Super-Resolution

Image Super-Resolution, SR refers to the technique of reconstructing or generating a **high-resolution image** from one or more **low-resolution images** of the same scene.

➤ Types of RSISR

◆ According to the input:

- **Single-Image Super-Resolution, SISR**
Reconstructing a high-resolution image from one low-resolution image.
- **Multi-Image / Multi-Frame Super-Resolution, MISR/MFSR**
Reconstructing a high-resolution image from multiple low-resolution images of the same scene, often with sub-pixel shifts.



Definition and Types of RSISR

➤ Types of RSISR

◆ According to whether deep learning is used:

- Traditional Methods
- Deep Learning Methods

◆ According to whether supervision is used:

- Supervised methods
- Unsupervised methods

◆ According to the modality of RSI for super-resolution:

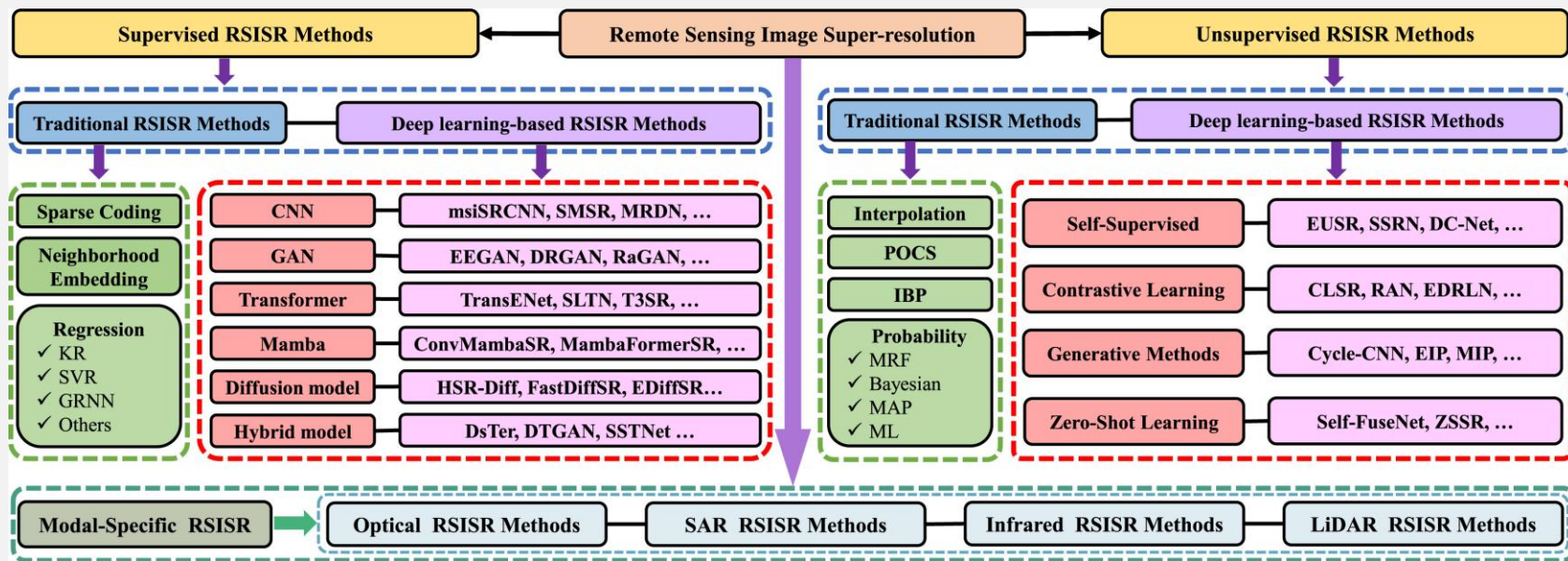
- | | |
|-------------------------|--------------------------|
| • Optical RSISR methods | • Infrared RSISR methods |
| • SAR RSISR methods | • LiDAR RSISR methods |



Definition and Types of RSISR

Types of RSISR

Qi et al. Advancing image super-resolution techniques in remote sensing: A comprehensive survey. ISPRS Journal of Photogrammetry and Remote Sensing, 2026

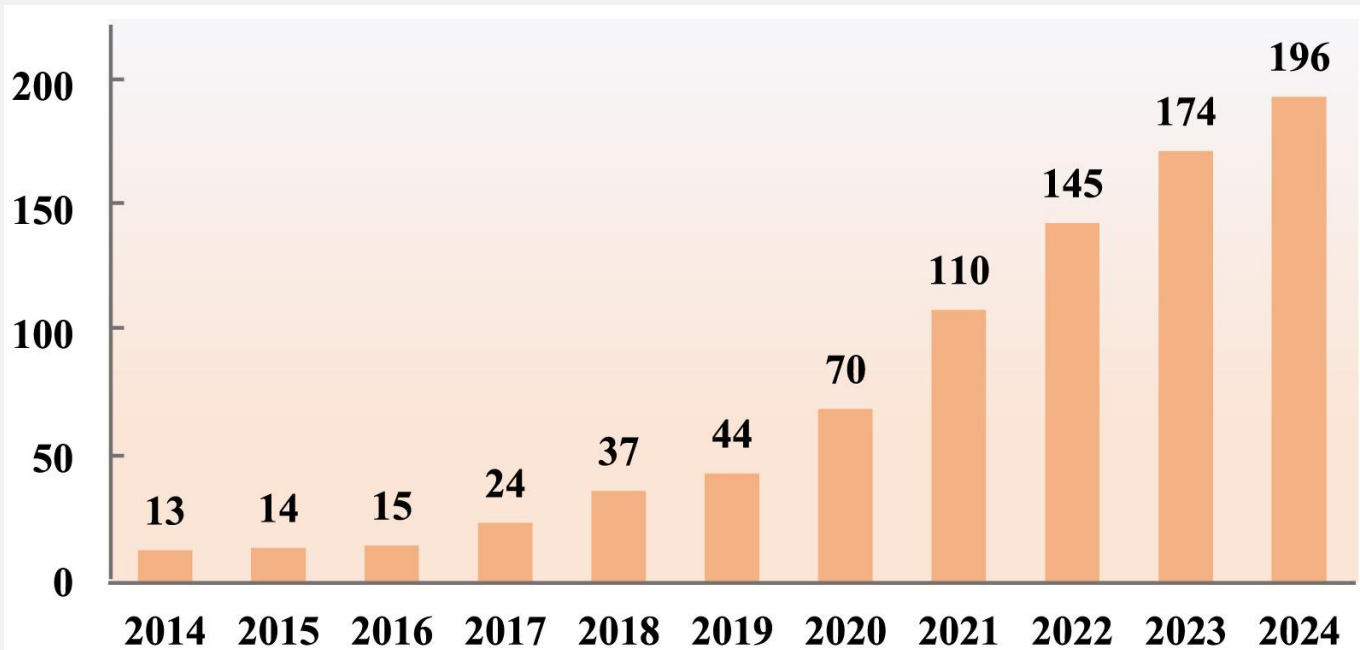


Taxonomy (分类系统) of the existing RSISR methods



Why Is RSISR Important?

Qi et al. Advancing image super-resolution techniques in remote sensing: A comprehensive survey. ISPRS Journal of Photogrammetry and Remote Sensing, 2026



The number of papers about SR algorithms for RSI since 2014
(according to Web of Science)



Why Is RSISR Important?

- Remote sensing image super-resolution is needed because many Earth observation applications require **finer spatial details** than the original satellite or airborne sensor can provide.

- **To overcome sensor resolution limitations**

- Remote sensing sensors are constrained by hardware, orbit altitude, cost, revisit frequency, and physical imaging conditions. High-resolution satellites are expensive and often have limited coverage or revisit time. Super-resolution provides a way to enhance image spatial resolution **without changing the sensor hardware**.

- **To obtain more detailed ground information**

Low-resolution remote sensing images often lose small-scale spatial details. Super-resolution can make objects and boundaries clearer, such as building, roads, rivers, vehicles, ships, urban structures.



Why Is RSISR Important?

- **To reduce mixed-pixel problems**
 - In medium- or low-resolution images, one pixel may contain several land-cover types, such as vegetation, soil, water, and buildings. This is known as the **mixed-pixel problem**. Super-resolution helps provide finer spatial information and can improve land-cover classification and object extraction.
- **To improve downstream remote sensing task**
 - Higher-resolution images can improve the performance of many remote sensing applications, including: land-use and land-cover classification, object detection, semantic segmentation, change detection, disaster assessment, urban monitoring, agricultural monitoring, environmental monitoring, etc.



Why Is RSISR Important?

- **To enhance the value of existing remote sensing data**
 - There are large archives of historical satellite images with relatively low spatial resolution. Super-resolution can improve their visual quality and analytical usability, making old data more valuable for long-term studies such as urban expansion, vegetation change, and climate-related analysis.
- **To balance spatial, spectral, and temporal resolution trade-offs**
 - Remote sensing systems usually face a trade-off:
 - high spatial resolution often means lower temporal coverage or fewer spectral bands;
 - high spectral resolution images, such as hyperspectral images, often have lower spatial resolution;
 - high temporal resolution sensors often have coarser spatial resolution.

Remote sensing image super-resolution is important because it enhances spatial detail beyond the original sensor resolution, thereby improving image interpretation, data usability, and the performance of downstream remote sensing applications.



Application Scenarios of RSI Super-Resolution

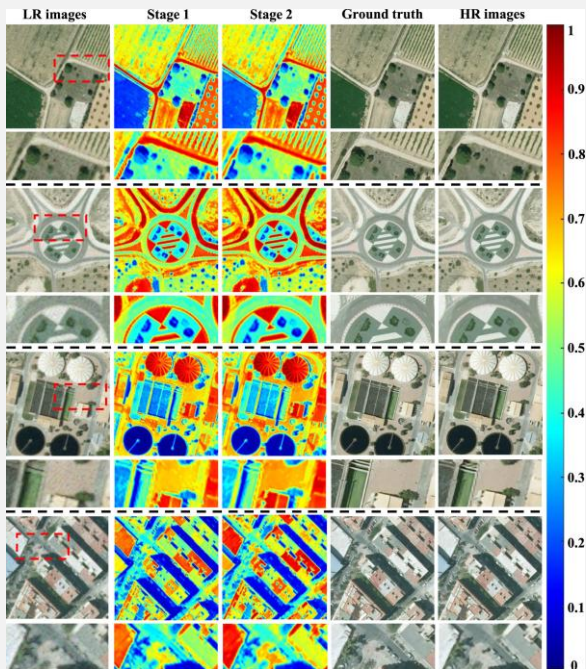
遥感影像超分辨率重建 (Remote Sensing Image Super-Resolution, **RSISR**) 的核心作用：在不改变传感器硬件条件的情况下，提高遥感影像的空间分辨率，从而增强目标细节表达、解译能力和下游任务性能。

➤ 应用场景1：细节增强

- 超分辨率可以增强影像中的纹理、边缘和形状信息，使小目标或边界更清晰，有助于自动识别与人工判读。

低分辨率影像中，小目标往往只有几个像元，甚至发生混叠；超分后可以提升：

- 目标可分辨性
- 边界完整性
- 纹理清晰度
- 检测与分割精度



Qi et al. Advancing image super-resolution techniques in remote sensing: A comprehensive survey. ISPRS Journal of Photogrammetry and Remote Sensing, 2026



Application Scenarios of RSI Super-Resolution

➤ 应用场景2：下游任务提升类

- 分类
- 目标检测
- 分割
- 变化检测

	Input	Label	UNet	Deeplab V3+	Swin-Transformer	Mask2former	ISFA_DLM	DSRL	SRAL	SRSeg(ours)
(1)										
		OA/mIoU/F1	73.06/0.4924/0.4460	74.99/0.5937/0.7041	83.73/0.7330/0.7927	85.86/0.7807/0.8464	86.88/0.8026/0.8597	88.56/0.8368/0.8725	91.32/0.8587/0.9069	93.38/0.8782/0.8534
(2)										
		OA/mIoU/F1	81.01/0.6400/0.6832	87.23/0.7625/0.8314	90.47/0.8192/0.8592	90.49/0.8160/0.8671	91.86/0.8267/0.8744	92.70/0.8481/0.8822	92.85/0.8679/0.8987	93.53/0.8762/0.9026
(3)										
		OA/mIoU/F1	70.74/0.7658/0.7495	75.86/0.7916/0.8101	91.07/0.8354/0.8865	89.14/0.8032/0.8687	91.62/0.8523/0.9002	91.36/0.8466/0.8936	92.26/0.8702/0.9126	93.19/0.8966/0.9455
(4)										
		OA/mIoU/F1	84.87/0.7034/0.7527	81.18/0.6788/0.7814	90.71/0.8527/0.8961	86.77/0.8039/0.8510	92.65/0.8767/0.9006	87.15/0.8185/0.8877	92.00/0.8649/0.8857	93.48/0.8953/0.9103
(5)										
		OA/mIoU/F1	66.98/0.5857/0.7063	75.57/0.6074/0.7560	85.06/0.7972/0.8081	88.20/0.8396/0.8442	91.09/0.8576/0.8607	92.11/0.8728/0.8683	92.41/0.8786/0.8810	93.35/0.8842/0.9097
(6)										
		OA/mIoU/F1	83.96/0.6826/0.5998	84.58/0.6639/0.6244	82.32/0.6986/0.8103	88.68/0.8542/0.8685	86.93/0.8334/0.8359	86.72/0.8486/0.8316	90.89/0.8735/0.8927	92.76/0.8936/0.9087

超分分类结果

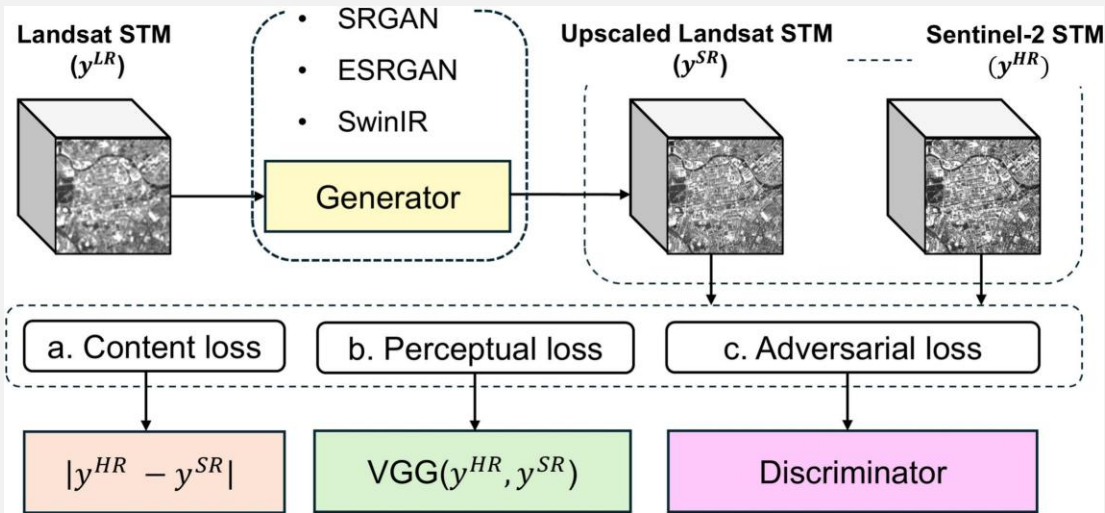


Application Scenarios of RSI Super-Resolution

➤ 应用场景3：专业监测类

- 农业
- 林业
- **城市**
- 灾害
- 军事

STM (**spectral-temporal metrics**) images are pixel-wise statistics (e.g., minimum, maximum, median, or quartiles) computed per band or spectral index from all available observations within a predefined time period



Overview of generative SR models for enhancing Landsat **STM** spatial resolution with Sentinel-2 STM used for training.

Pham et al. Mapping urban built-up types from 2000 to 2022 at 10-m resolution using super-resolution of Landsat spectral-temporal metrics and center-patch classification. RSE, 2026

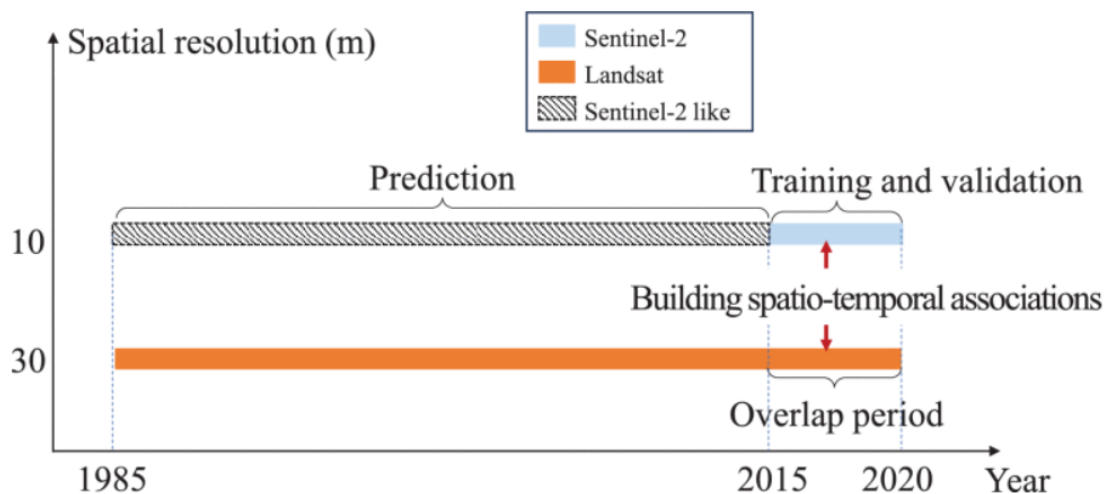


Application Scenarios of RSI Super-Resolution

➤ 应用场景4：数据融合与预处理

- 多源分辨率匹配
- 高光谱空间增强
- **历史数据再利用**

Purpose of our method: establishing spatiotemporal associations using the overlapping time periods of Landsat and Sentinel-2 imagery to predict the satellite dynamic changes from 1985 to 2015.



Xiong, et al. Retrieving Historical Images at 10-m Resolution From 1985 to 2015 Through STARS-Net: A Spatiotemporal Attention Referenced Super-Resolution Network. IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.2025



Open Source Dataset for RSI Super-Resolution

Name	Year	Data source	Numbers	Resolution
UCMD	2010	Air-Spaceborne	2100	256 × 256
WHU-RS19	2012	Google Earth	1005	600 × 600
RSSCN7	2015	Google Earth	2800	400 × 400
AID	2016	Aerial	10,000	400 × 400
KOSD	2016	Draper Satellite	1620	3099 × 2329
SIRI-WHU	2016	Google Earth	2400	200 × 200
NWPU-RESISC45	2017	Google Earth	31,500	256 × 256
PatternNet	2018	Google Earth	30,400	256 × 256
DOTA	2018	JL-1, GF-2	2806	4000 × 4000
SpaceNet2	2018	WV-2	10,595	162 × 162
DFC2019	2019	WV-3	2783	1024 × 1024
OPTIMAL-31	2019	Google Earth	1860	256 × 256
RSI-CB	2020	Crowdsourcing	60,000	128 × 128
				256 × 256

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Traditional Methods for RSISR

主要包括**插值法**、**基于稀疏编码法**、**基于回归法**、**全色税化法等**

➤ **插值法 (Interpolation)**

代表算法：最近邻插值、双线性插值、**双三次插值**(Cubic Convolution)。
通过已知像素点的几何位置关系，直接估计未知像素的灰度值。

双三次卷积插值核心思想：

- 插值新像素值，**不取最近点、不取平均**
- 以待求像素为中心，选取周围 **4 行 4 列共 16 个原始像素**
- 用**三次多项式卷积核**计算每个像素权重
- 加权求和得到新像素灰度值

标准三次卷积基函数：

$$W(x) = \begin{cases} 1 - 2|x|^2 + |x|^3, & 0 \leq |x| < 1 \\ 4 - 8|x| + 5|x|^2 - |x|^3, & 1 \leq |x| < 2 \\ 0, & |x| \geq 2 \end{cases}$$

x: 目标点与原始像素的**距离偏移量**
距离越近权重越大，超出 2 个像素权重直接为 0

局限性：本质是信号重采样，无真实信息恢复，图像边缘易模糊或出现锯齿，高频细节丢失严重



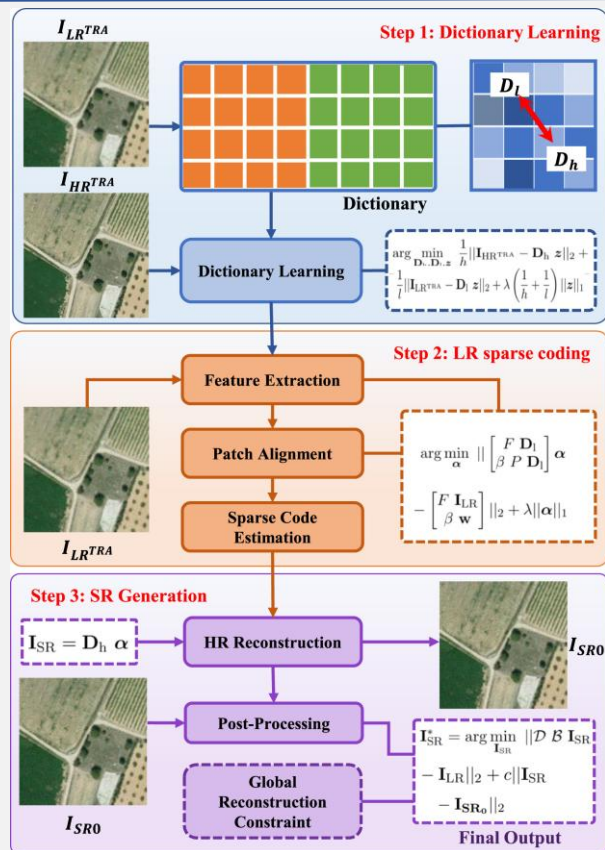
Traditional Methods for RSISR

➤ 基于稀疏编码法 Sparse coding-based methods

It consists of three algorithmic steps: dictionary learning, sparse coding and HR reconstruction.

- **Dictionary Learning:** 从配对的低分辨率与高分辨率训练图像块中学习 LR/HR 字典，建立两者之间的稀疏表示对应关系。
- **Sparse Coding:** 对输入低分辨率图像块提取特征，并在低分辨率字典上求解其稀疏系数表示。
- **HR Reconstruction:** 利用得到的稀疏系数在高分辨率字典中重建对应的高分辨率图像块，并融合生成最终超分辨率图像。

Qi et al. Advancing image super-resolution techniques in remote sensing: A comprehensive survey. ISPRS Journal of Photogrammetry and Remote Sensing, 2026

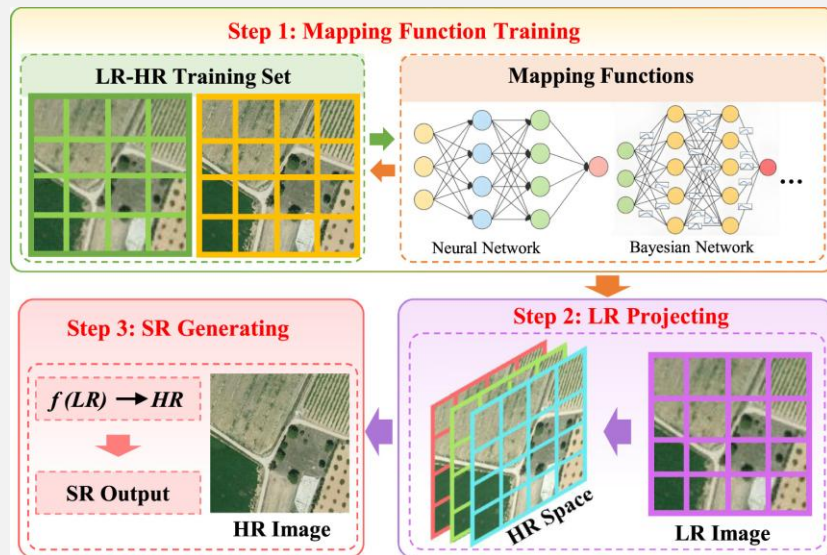




Traditional Methods for RSISR

➤ 基于回归法

- The regression-based RSISR methods learn the function mappings between LR and HR images, which regards super-resolution as a regression problem.
- More specifically, these methods usually extract features from LR images based on paired datasets to train regression models, and then use the trained models to predict and reconstruct HR images.
- Typically, regression-based methods consist of **three main steps: regression model training, projecting of LR images to HR image space, and final super-resolution result generation**



Including Kernel Regression, Support Vector Regression (SVR), General Regression Neural Network (GRNN), Others.

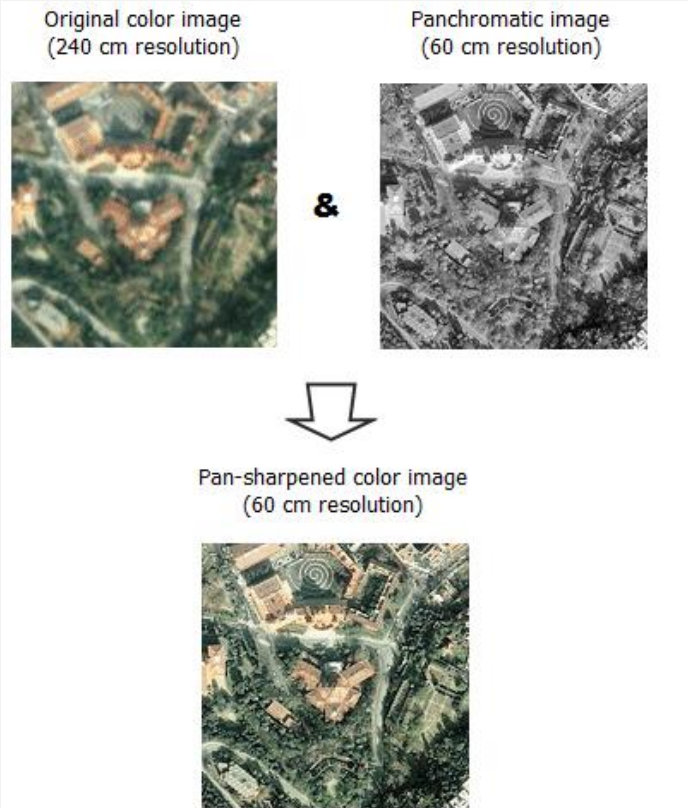


Traditional Methods for RSISR

➤ 全色锐化 Pan Sharpening

Pan sharpening uses a higher-resolution panchromatic image (or raster band) to fuse with a lower-resolution multiband raster dataset. The result produces a multiband raster dataset with the resolution of the panchromatic raster, where the two rasters fully overlap.

可用于创建全色锐化影像的影像融合方法：**Brovey** 变换；**Esri** 全色锐化变换；**Gram-Schmidt** 光谱锐化方法；**强度、色调、饱和度 (IHS)** 变换；以及**简单均值**变换。





Traditional Methods for RSISR

–Pan Sharpening 全色锐化

◆ Brovey 变换

将各个重采样的多光谱像素乘以相应全色像素亮度与所有多光谱亮度总和的比值。
假定全色图像所跨越的光谱范围与多光谱通道覆盖的范围相同。

- 常规使用三个Red、Green、Bblue三个多光谱波段时，Brovey变换公式为：

$$\text{Red_out} = \text{Red_in} / [(\text{Blue_in} + \text{Green_in} + \text{Red_in}) * \text{Pan}]$$

- 使用权重的R、G、B、NIR四个多光谱波段时，Brovey变换公式为：

$$\text{DNF} = (P - IW * I) / (RW * R + GW * G + BW * B)$$

$$\text{Red_out} = R * \text{DNF}$$

$$\text{Green_out} = G * \text{DNF}$$

$$\text{Blue_out} = B * \text{DNF}$$

$$\text{Infrared_out} = I * \text{DNF}$$

P = panchromatic image

R = red band

G = green band

B = blue band

I = near infrared

W = weight



Traditional Methods for RSISR

-Pan Sharpening 全色锐化

◆ Esri 全色锐化

Esri 全色锐化变换使用加权平均和附加的近红外波段（可选）来创建其全色锐化输出波段。**加权平均WA**的结果可用来创建**调整值 (ADJ)**，随后将使用该值计算输出值。

WA 可以看作是由多光谱各波段按一定权重合成出来的“近似亮度图像”

$$WA = W_R R + W_G G + W_B B, \quad W_R + W_G + W_B = 1$$

$$ADJ = \text{pan image} - WA$$

$$\text{Red}_{\text{out}} = R + ADJ$$

$$\text{Green}_{\text{out}} = G + ADJ$$

$$\text{Blue}_{\text{out}} = B + ADJ$$

$$\text{Near_Infrared}_{\text{out}} = I + ADJ$$

- 多光谱波段的权重取决于多光谱波段的光谱灵敏度曲线与全色波段的重叠程度。
- WA权重是相对的，将在使用时进行归一化。
- 与全色波段重叠程度最大的多光谱波段应获得最大的权重值。与全色波段完全不重叠的多光谱波段应获得权重值 0。



Traditional Methods for RSISR

-Pan Sharpening 全色锐化

◆ Gram-Schmidt

Gram-Schmidt 全色锐化方法核心是用向量正交化分离空间与光谱信息，用高分辨率全色 (Pan) 替换低分辨率模拟全色，再逆变换重建高分辨率多光谱 (MS) 影像。

- 按传感器光谱响应加权 MS 波段，生成与 PAN 光谱范围匹配的低分辨率全色影像 S 。
- 视每个波段为高维向量（维度 = 像素数），以 S 为第一分量 C_1 ，逐次 Gram-Schmidt 正交化其余 MS 波段，得到 **互不相关**的正交分量 $C_1, C_2, \dots, C_n$ 。
- 将正交分量 C_1 （低分辨率 S ）替换为 **高分辨率 PAN**，并做直方图匹配以保持统计一致性。
- 用替换后的分量作 Gram-Schmidt 逆变换，重建 **高分辨率多光谱影像**，空间分辨率与 PAN 一致，光谱特征与原始 MS 接近。

一些常用传感器的建议权重如下（分别为红色、绿色、蓝色和的近红外）

- GeoEye - 0.6、0.85、0.75、0.3
- IKONOS - 0.85、0.65、0.35、0.9
- QuickBird - 0.85、0.7、0.35、1.0
- WorldView-2 - 0.95、0.7、0.5、1.0



Traditional Methods for RSISR

-Pan Sharpening 全色锐化

◆ IHS Intensity-Hue-Saturation, 亮度-色调-饱和度) 色彩空间变换

核心思路:

- 多光谱影像MS转为IHS 空间, 分离亮度 I、色调 H、饱和度 S;
 - 色调 H: 地物颜色属性 (光谱信息, 必须保留)
 - 饱和度 S: 色彩鲜艳程度 (光谱信息, 尽量保留)
 - 亮度 I: 明暗 / 空间纹理信息 (用高分辨率 PAN 替换)
- 用高分辨率全色影像 PAN替换原亮度分量, 再逆变换转回 RGB, 实现空间分辨率提升, 保留地物色彩特征

1. 正变换
(RGB →
IHS)

$$\begin{cases} I = \frac{R + G + B}{3} \\ H = \arctan \left(\frac{\sqrt{3}(G - B)}{2R - G - B} \right) \\ S = 1 - \frac{3 \min(R, G, B)}{R + G + B} \end{cases}$$

2. 将低分辨率亮度I, 替换为直方图匹配后的高分辨率全色 PAN:

$$I_{new} = PAN_{matched}$$

3. 利用新亮度 I_{new} + 原、反推高分辨率多光谱



Traditional Methods for RSISR— Weaknesses

- In summary, traditional super-resolution research work mainly focused on the period before 2016, which was attributed to the fact that deep learning methods had not been fully explored before that time.
- Generally speaking, the calculation of conventional supervised SR methods is relatively simple and suitable for application scenarios with limited hardware resources.
- Nevertheless, they rely heavily on the relationship between local pixels, which cannot fully extract deep features and high-level semantic information of images, resulting in suboptimal performance.
- In addition, in high super-resolution (such as 4 or 8) reconstruction tasks, these conventional methods often fail to recover sufficient details, resulting in blurry or distorted results. *For example, regression-based methods are easily affected by noise, resulting in a decrease in reconstruction quality.*



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Deep Learning Methods for RSISR

➤ CNN-based RSISR

- SRCNN: 图像超分领域的开创性卷积神经网络
- msSRCNN: 第一个基于CNN的RSISR模型
- ESPCN: 引入高效亚像素卷积实现端到端上采样
- EDSR, RCAN, RNAN, HAN, HANNet, HSENet等

➤ Transformer-based RSISR

- SwinIR: 基于窗口自注意力机制的经典超分模型
- HAT: 采用混合注意力机制优化特征提取能力
- TTST: 针对遥感图像特性设计的Transformer模型

➤ GAN-based RSISR

➤ Mamba-based RSISR

➤ Diffusion model-based RSISR

➤ Hybrid RSISR methods



Deep Learning Methods for RSISR

类型	模型	提出者	时间	主要特性
CNN-based	SRCNN	Dong et al ^[1-2]	2014,2016	深度学习图像超分辨率领域的开山之作
	msiSRCNN	Liebel et al ^[3]	2016	第一个RSISR模型，基于SRCNN
	ESPCN	Shi et al ^[4]	2016	引入高效亚像素卷积实现端到端上采样
	EDSR	Lim et al ^[31]	2017	移除批量归一化层、深层连接、全局残差学习
	RCAN	Zhang et al ^[35]	2018	深度残差结构、通道注意力机制
	RNAN	Zhang et al ^[63]	2019	残差非局部注意力机制
	HAN	Niu et al ^[37]	2020	层次注意模块、通道-空间注意模块
	HAUNet	Wang et al ^[38]	2023	基于混合卷积的注意力机制
Transformer-based	HSENet	Lei et al ^[39]	2023	混合尺度自相似性开发模块
	SwinIR	Liang et al ^[44]	2021	结合CNN局部特征提取能力与Transformer全局特征建模优势
	TTST	Xiao et al ^[57]	2024	Top-k Token可选择注意力机制
	HAT	Chen et al ^[41]	2023	结合通道注意力与窗口自注意力、重叠交叉注意力模块



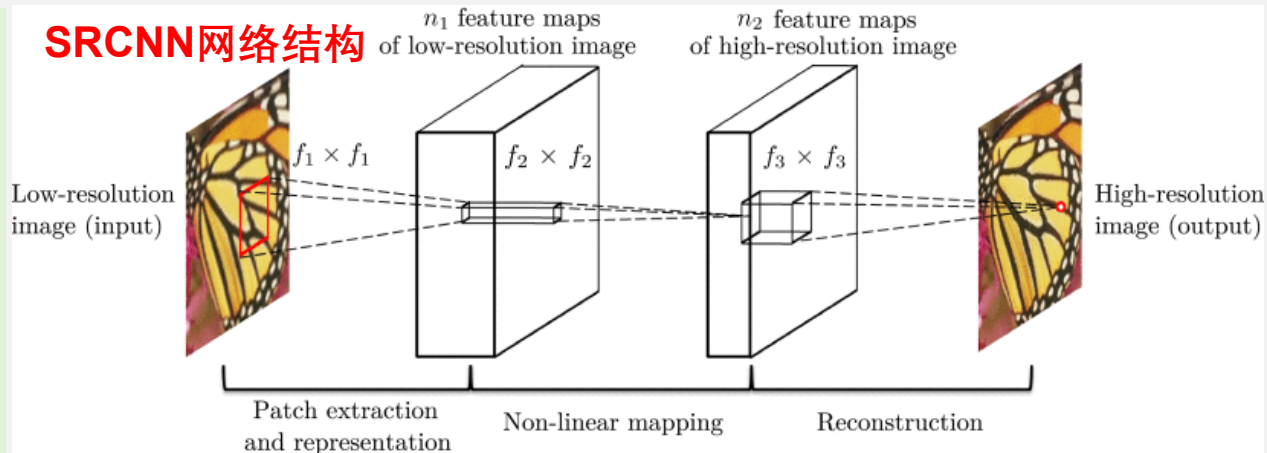
Deep Learning Methods for RSISR: CNN-Based

◆ SRCNN (Super-Resolution CNN)

于2014提出，并于2016作微调，是深度学习图像超分辨率领域的开山之作，奠定了基于CNN的单图像超分辨率研究范式

- Dong, C., et al. *Image Super-Resolution Using Deep Convolutional Networks*. *Proc. Eur. Conf. Comput. Vis.*, 2014.
- Dong, C., et al.. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2016.

- ✓ 图像块提取：对LR图像插值放大后，利用卷积层提取浅层特征；
- ✓ 非线性映射：通过卷积层将LR特征非线性变换至HR特征空间；
- ✓ 图像重建：再次卷积聚合特征，生成最终HR图像



优点： 开创性构建了深度学习超分基线，结构简洁且效果显著优于传统SR方法；

缺点： 计算效率偏低（基于放大后HR特征图运算）、感受野较小且依赖预处理插值。



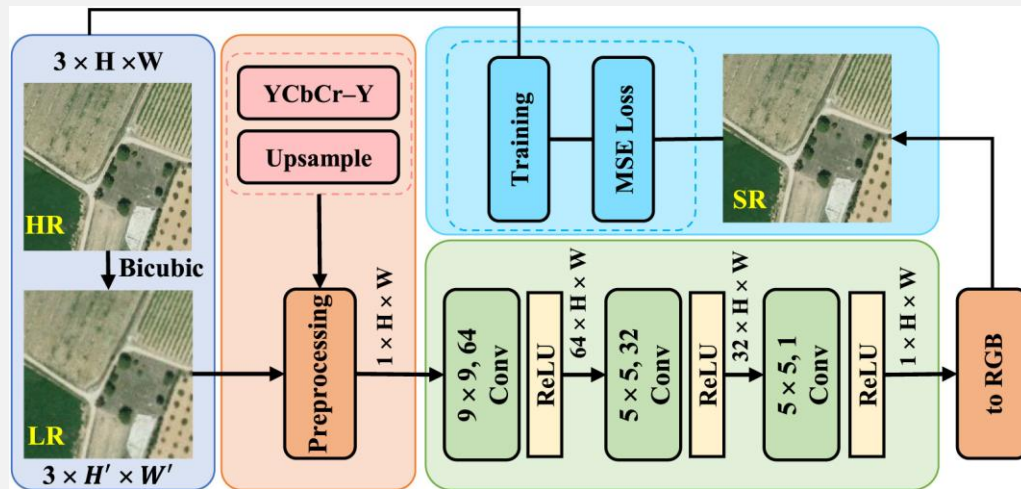
Deep Learning Methods for RSISR: CNN-Based

◆ **msiSRCNN**

Liebel, L. & Korner, M. Single-image super resolution for multispectral remote sensing data using convolutional neural networks. *Int. Arch. Photogramm. Remote. Sens. Spat. Inf. Sci.*, 2016.

The first CNN-based RSISR method which utilizes remote sensing data (Sentinel-2 images) to train and fine-tune SRCNN.

Fine-tune SRCNN based on the luminance component of the YCbCr color space of RSIs and performed performance evaluation.



The schematic diagram of the SRCNN-based RSISR method

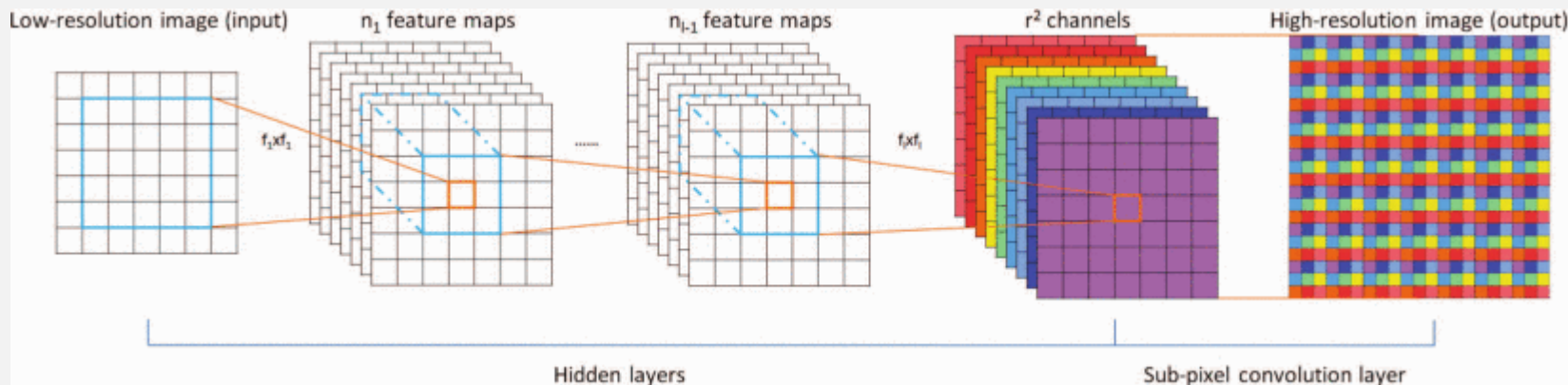


Deep Learning Methods for RSISR: CNN-Based

◆ ESPCN: Efficient Sub-Pixel CNN

Shi, W.Z, et al. Real-Time Single Image and Video Super-Resolution Using an Efficient Sub-Pixel Convolutional Neural Network. CVPR 2016.

- 核心理念：**提出Sub-pixel Convolution（亚像素卷积层），彻底颠覆了传统超分任务中“先插值放大、后卷积处理”的流程。该网络将所有卷积运算全部在LR特征图上完成，最后通过亚像素卷积层的像素重排机制，将LR特征图直接高效映射为HR图像，极大降低了计算复杂度。



The proposed ESPCN, with two convolution layers for feature maps extraction, and a sub-pixel convolution layer that aggregates the feature maps from LR space and builds the SR image in a single step.

优点：推理速度极快；在相同放大倍数下重建效果显著优于SRCNN。

缺点：网络结构相对简单，感受野范围较小，对复杂纹理细节的恢复能力与深层网络相比存在差距，在大尺度超分场景下表现受限。



Lect.13 Super-Resolution for Remote Sensing Image (RSI)

- Review of RSI Super-Resolution**
- Traditional Methods for RSI Super-Resolution**
- Deep Learning Methods for RSI Super-Resolution**
- Performance Evaluation Methods**
- Future Prospects**
- Cases**



Performance Evaluation Methods

➤ Qualitative evaluation

- ◆ Visual Comparison
- ◆ Expert visual evaluation
- ◆ Application-guided evaluation

➤ Quantitative evaluation

- ◆ MSE均方误差(mean square area)
- ◆ RMSE均方根误差(root mean square area)
- ◆ PSNR峰值信噪比 peak signal-to-noise ratio)
- ◆ SSIM 结构相似性 (structural similarity)
- ◆ LPIPS感知图像块相似性 (learned perceptual image patch similarity)

徐一帆等.图像超分辨率方法研究综述.计算机工程与应用,2026.



Lect.13 Super-Resolution for Remote Sensing Image (RSI)

- Review of RSI Super-Resolution
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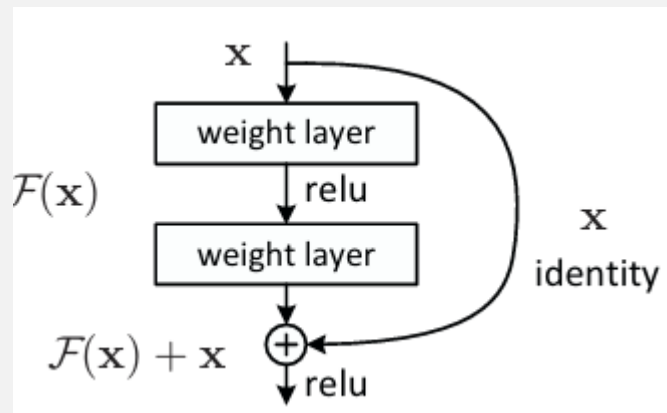
Future Prospects

- Real-world degradation modeling in RSISR
- Foundation models and transfer learning in RSISR
- Multi-modal RSISR methods
- Better evaluation metrics and benchmarks
- Diffusion-Mamba based RSISR methods



ResNet-34

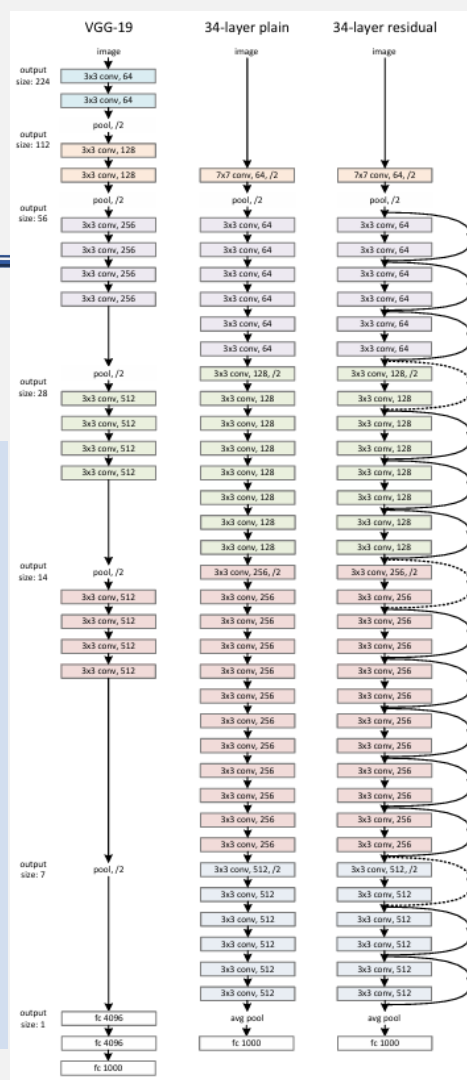
He, K., Zhang, X., Ren, S., & Sun, J. (2016). **Deep residual learning for image recognition**. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.



Residual learning: a building block

Example network architectures for ImageNet.

- Left: the VGG-19 model [41] (19.6 billion FLOPs) as a reference.
- Middle: a plain network with 34 parameter layers (3.6 billion FLOPs).
- Right: a residual network with 34 parameter layers (3.6 billion FLOPs).
- The dotted shortcuts increase dimensions.





Cases

- Single remote sensing image super-Resolution via a generative adversarial network with stratified dense sampling and chain training
- Mapping urban built-up types from 2000 to 2022 at 10-m resolution using super-resolution of Landsat spectral-temporal metrics and center-patch classification
- SeG-SR: Integrating Semantic Knowledge Into Remote Sensing Image Super-Resolution via Vision-Language Model
- Retrieving Historical Images at 10-m Resolution From 1985 to 2015 Through STARS-Net: A Spatiotemporal Attention Referenced Super-Resolution Network



References

- ArcGIS help: <https://pro.arcgis.com/en/pro-app/latest/tool-reference/geoanalytics-desktop/an-overview-of-the-geoanalytics-desktop-toolbox.htm>
- Dong, C., et al. Image Super-Resolution Using Deep Convolutional Networks. Proc. Eur. Conf. Comput. Vis., 2014.
- Dong, C., et al.. Image Super-Resolution Using Deep Convolutional Networks. IEEE Transactions on Pattern Analysis and Machine Intelligence. 2016.
- Shi, W.Z, et al. Real-Time Single Image and Video Super-Resolution Using an Efficient Sub-Pixel Convolutional Neural Network. CVPR 2016.
- Liebel, L. & Korner, M. Single-image super resolution for multispectral remote sensing data using convolutional neural networks. Int. Arch. Photogramm. Remote. Sens. Spat. Inf. Sci., 2016.
- Qi et al. Advancing image super-resolution techniques in remote sensing: A comprehensive survey. ISPRS Journal of Photogrammetry and Remote Sensing, 2026
- Other relevant papers.



Reading , Exercises & Lab

1. Read reference 1 and 2, as well as one relevant SCI paper.
2. What is geospatial big data? Which sources can geospatial big data be obtained? Which platforms can geospatial big data be downloaded?
3. Describe at least three application cases of geospatial big data analysis.
4. Do Lab 12.

ArcGIS Pro 以RestNet-34为缺省设置的Backbone